

Enabling AI-Assisted Pharmacometric Workflows Using Synthetic Data

Andrew Stein¹, Alex Pogodaev²

1. Novartis, Cambridge, MA, USA – andrew.stein@novartis.com
2. Novartis, Basel, Switzerland

This work is sponsored by Novartis Pharma AG
Poster presented at ACOP 2026, held on October 11-14, 2026

KEY FINDINGS & CONCLUSIONS

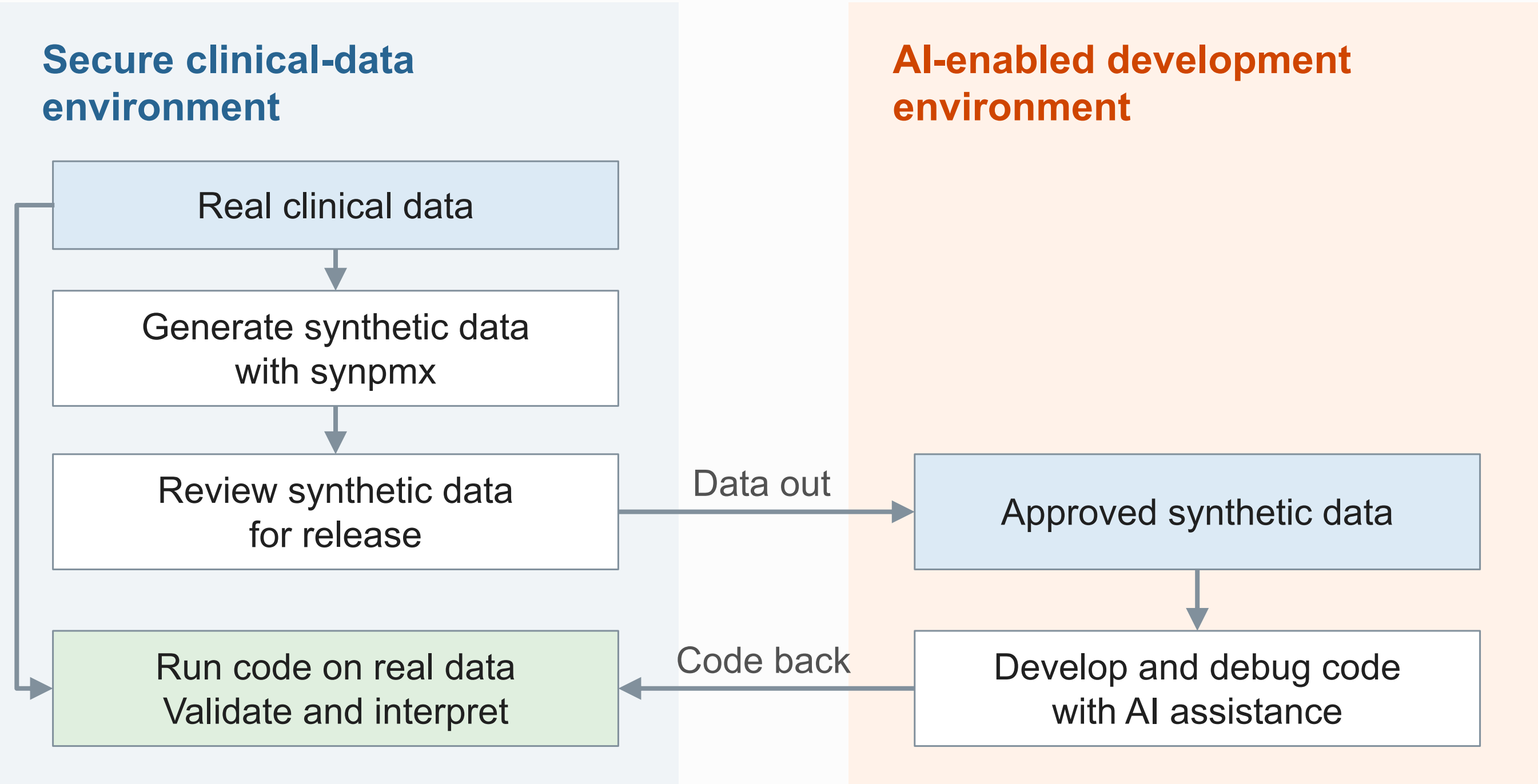
- synpmx is a new R package for generating synthetic data from real datasets. It is available at: iamstein.github.io/synpmx
- Intended use case: transferring synthetic data out of secure computing environment to enable AI-assisted code generation in an AI-enabled development environment.
- Further work is needed to develop safeguards against membership inference from reported model parameters.



OBJECTIVE

synpmx generates synthetic pharmacometric datasets for artificial intelligence (AI)-assisted code development when clinical data must remain in a secure computing environment. Synthetic data provides the columns, dosing records and observations needed to develop exploration, model-fitting and diagnostic code.

WORKFLOW



SELECTED METHODS

Function	Generation and specification	Formal Privacy	Reportable Fingerprint
synpmx_prior()	Simulate a public model and trial design. Requires a public model and study design.	Yes	Yes
synpmx_model()	Fit longitudinal pharmacokinetic (PK) and pharmacodynamic (PD) models. Requires column roles and nominal times.	No	Yes
synpmx_avatar()	Blend similar patient profiles, inspired by AVATAR [1]. Requires column roles. Nominal times recommended.	No	No

A reportable fingerprint is a compact model-and-parameter representation that generates synthetic data without retaining source patient profiles.

Deep learning methods [2] are outside scope because learned weights lack a compact, interpretable fingerprint and methods are difficult to explain to those responsible for data privacy. Other methods provide differential privacy, but noise reduced fidelity in the small-study settings.

FITTED-MODEL GENERATOR

synpmx_model()

Component	Description
Disclosure safeguards	Generate new identifiers. Drop columns without defined roles and cohorts with fewer than 3 patients. Exclude categorical levels held by fewer than 3 patients and sample remaining levels. Threshold (n) for cohorts and categorical levels are configurable.
Dosing	Estimate constant per-cycle probabilities within each cohort for skipped cycles, dose reductions and discontinuation. Simulate schedules from these probabilities.
Observations	Estimate attendance probabilities within each cohort, endpoint and nominal visit. Simulate missed observations.
Population PK	Fit two compartment model with nlmixr2 first. Fall back to one compartment model if convergence or generation checks fail. Users have the option to specify a model.
PD	For continuous observations: Fit constant, linear or exponential time courses with subject variability and residual error. Pool arms by default. Use pd_by_arm = TRUE to fit each arm separately. For discrete observations: sample by arm and by nominal visit.
Limits of Quantification (LOQ)	Apply declared censoring limits. Otherwise, for predominantly positive continuous endpoints, use half the smallest positive observation as a heuristic lower bound.

WORKED EXAMPLE

Public xgxr::mad simulated data combines PK concentrations with continuous, ordinal, count and binary PD endpoints. PK and continuous PD are shown. Blue shows source data and orange shows synthetic data.

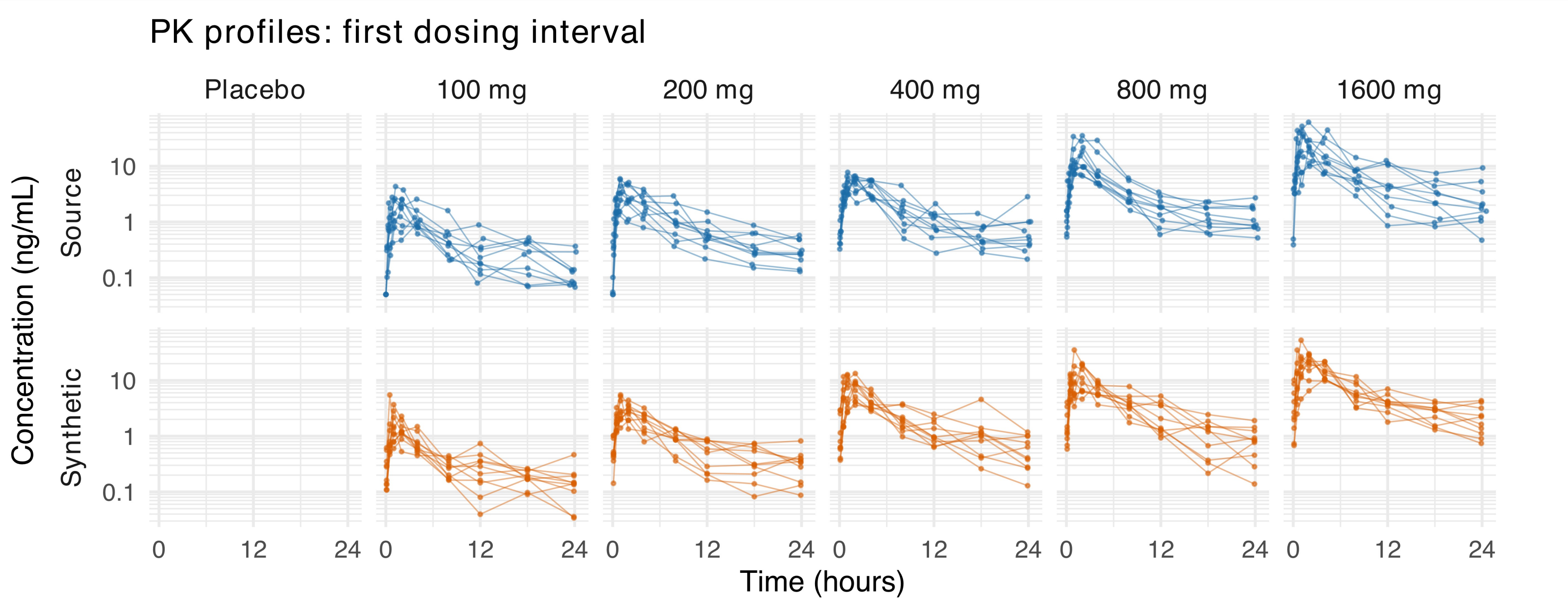


Figure 3a. The two-compartment model reproduces the peak and biphasic decline over the first dosing interval.

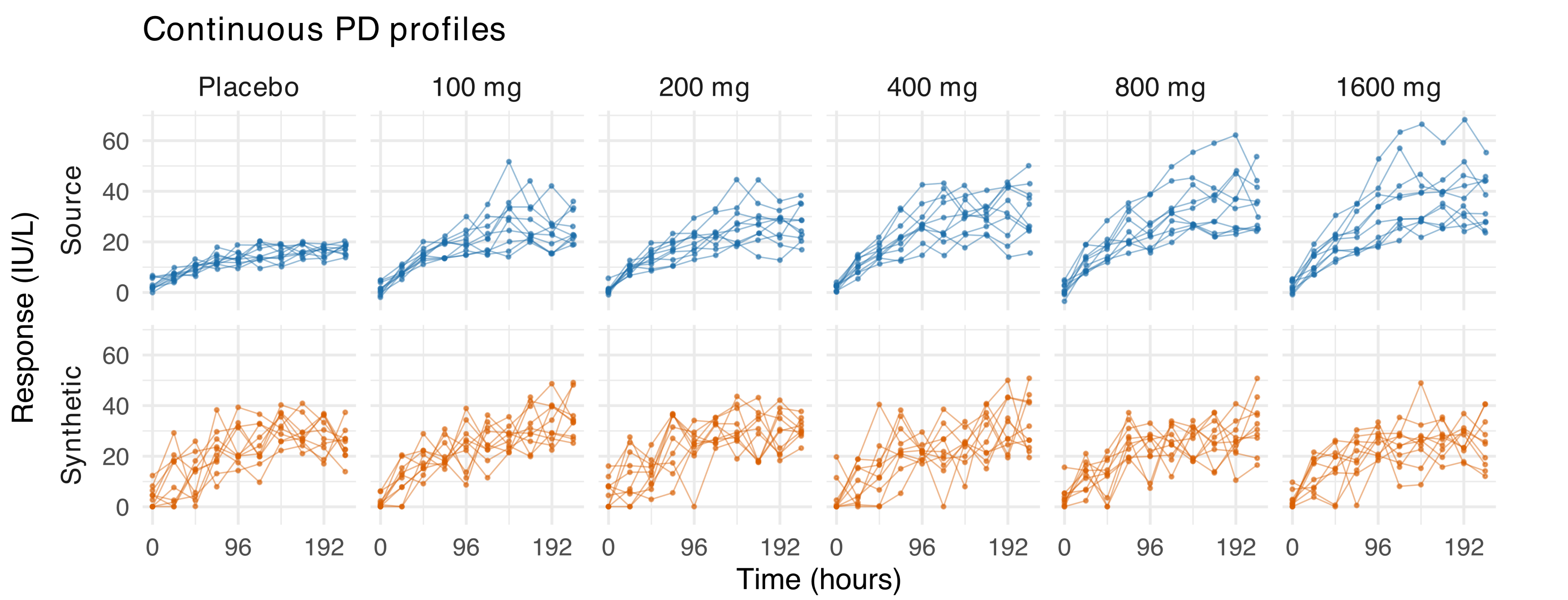


Figure 3b. Pooling removes between-arm response differences. Use pd_by_arm = TRUE to fit separate time courses by arm.

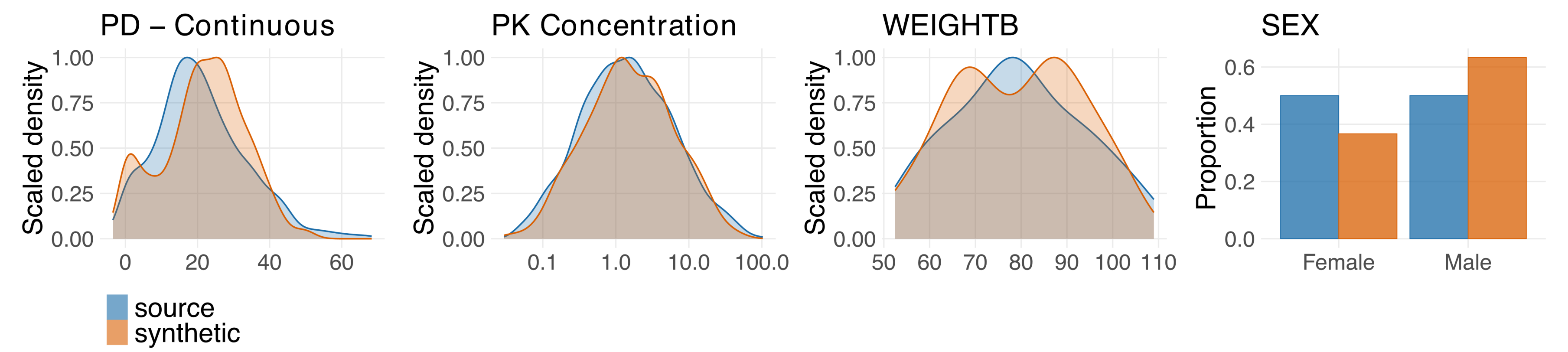


Figure 3c. Comparison shows similar distributions for real and synthetic data. A scorecard is also used to check synthetic data structure, distribution and disclosure of rare patient characteristics iamstein.github.io/synpmx/articles/scorecard.html

DATASETS EVALUATED

Dataset	Patients	Design feature
case1_pkpd	180 (simulated)	Multiple arms; censored concentrations
mad	60 (simulated)	Multiple doses; mixed endpoint types
warfarin	32 (actual)	Single oral dose; delayed response
wbcSim	45 (simulated)	Infusions; white-cell nadir and recovery
mavoglurant	120 (actual)	Repeated occasions;
theo_md	12 (actual + simulated)	Repeated oral doses; small cohort
nimoData	12 (actual)	Weekly infusions; small cohort
pheno_sd	59 (actual)	Neonatal care; sparse, irregular sampling
mixroute_sim	90 (simulated)	Intravenous and subcutaneous dosing
onc_sim	200 (simulated)	Trough sampling; tumor response; dose changes

AVAILABILITY AND DISCLOSURES

R package prototype, source code and worked examples: iamstein.github.io/synpmx/

Internal discussions on use are ongoing. Further work is needed to develop safeguards against membership inference from reported model parameters.

Package written by ChatGPT and Claude; not yet formally validated. Establishing criteria for validation is a next step. The authors reviewed and edited content and take full responsibility for this publication.

REFERENCES

- Destere A, Lombardi R, Labriffe M, et al. Can synthetic data overcome the privacy and fidelity bottleneck in Pharmacometrics? A comparative benchmark using a daptomycin population pharmacokinetic model. medRxiv [preprint]. 2026. doi:10.64898/2026.05.30.26354512.
- Jiang Y, García-Durán A, Bachali Losada I, Girard P, Terranova N. Generative models for synthetic data generation: application to pharmacokinetic/pharmacodynamic data. J Pharmacokinet Pharmacodyn. 2024;51:877–885. doi:10.1007/s10928-024-09935-6.